ModelIt!

Student Activity Pack

Telephone Game — How Signals Travel

Why Messages Get Lost Along the Way

Grade NE | NGSS MS-LS1-8, MS-PS4-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Have you ever played the telephone game? What happens to the message by the end?

2 Why do you think phone calls sometimes sound crackly or break up?

3 If you could control a robot with your brain, what would you make it do?

4 Draw what you think a brain signal looks like as it travels to your hand.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Signal Strength | Distance Traveled | Noise Level | Message Accuracy

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Strong and Close

High signal, short distance, low noise

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Weak and Far

Low signal, far distance, high noise

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Noisy Room

High signal, short distance, high noise

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Shielding — Protection around the signal path that blocks outside interference

Signal Encoding — How the message is packaged — better encoding resists noise

Number of Channels — How many signal paths are used — more channels means more data

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why is it so hard for your brain to control a machine on the other side of the room? Use evidence from your simulation to support your answers.

1. **Why does playing telephone with more people make the message less accurate?**

2. **Which factor hurt accuracy the most in your model — distance, noise, or low signal?**

3. **How do real engineers solve the noise problem in electronic systems?**

4. **Why is it harder to read brain signals than to read a simple button press?**

5. **How did modeling three inputs affecting one output help you understand this system?**

8 STEM Challenge: Build a Signal Circuit

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YOUR ENGINEERING MISSION

Using simple circuits, build a signal chain where pressing a button at one end lights an LED at the other. Then test what happens when you extend the wire and add noise.

THE SCENARIO

A medical device company wants to build a system that reads a patient's muscle signals and moves a robotic arm. Your team needs to understand the challenges of sending signals over distance with noise. Build a circuit to test these limits!

Engineering Design Thinking

1. HOW WILL YOU MEASURE SIGNAL ACCURACY — BRIGHTNESS, SPEED, OR ON/OFF RELIABILITY?

2. WHAT LENGTH OF WIRE CAUSES THE SIGNAL TO NOTICEABLY WEAKEN?

3. CAN YOU SHIELD YOUR CIRCUIT FROM NOISE? HOW?

4. HOW COULD YOU BOOST THE SIGNAL TO OVERCOME DISTANCE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Biomedical Engineers design medical devices including brain-computer interfaces that help paralyzed patients control computers and robotic limbs with their thoughts. They earn $75,000–$130,000/year.